The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Residual Law
from a Centered Hidden-Response
Residual-Source Law

Abstract

The current active-primary theorem already proves that the current hidden-response Hessian minimization law is the canonical least-squares pullback of a smaller centered hidden-response residual law on the fixed displacement fibers. The next honest benchmark-facing move is therefore no longer another replay of that centered residual law, of the hidden-response Hessian minimization package, of the square-root operator core, of the normalized-lift package, of the body-level forcing law, or of the visible-response structures that follow from them. It is to derive or justify the centered residual law itself from still more primitive explicit substrate structure.

The present note takes that step in the cleanest quotient-energy form currently available below the centered-residual-law frontier. For each sector, it introduces a benchmark-faithful centered hidden-response residual-source law: the fixed proto-section benchmark base and fixed section-centered displacement body together with a finite-dimensional Euclidean residual-source carrier, a smooth surjective source-state projection onto the current residual state space, a smooth surjective source displacement readout to the actual displacement fibers, and a smooth positive self-adjoint residual-source response operator. The current centered residual law is then no longer primitive. It is canonically induced by the unique minimal-source-energy lift

$$J_\sigma^{\mathrm{disp,hid,res,src}}(y) = K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1} Q_\sigma^{\mathrm{disp,hid,res,src}}(y)^* \cdot \left(Q_\sigma^{\mathrm{disp,hid,res,src}}(y) K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1} Q_\sigma^{\mathrm{disp,hid,res,src}}(y)^* \right)^{-1},$$

the descended residual readout

$$\Pi_\sigma^{\mathrm{disp,hid,res}}(y) Q_\sigma^{\mathrm{disp,hid,res,src}}(y) = \Theta_\sigma^{\mathrm{disp,hid,res,src}}(y),$$

and the induced residual operator

$$L_\sigma^{\mathrm{disp,hid,res}}(y) = K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{1/2} J_\sigma^{\mathrm{disp,hid,res,src}}(y).$$

Equivalently,

$$\frac{1}{2} \left\| L_\sigma^{\mathrm{disp,hid,res}}(y) u \right\|^2 = \min_{Q_\sigma^{\mathrm{disp,hid,res,src}}(y) w = u} \frac{1}{2} \left\langle w, K_\sigma^{\mathrm{disp,hid,res,src}}(y) w \right\rangle,$$

so the current centered residual law is exactly the quotient minimal-source-energy law of that smaller explicit source package.

The benchmark boundary remains unchanged throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. Once this smaller residual-source law is fixed, the same centered residual law, the same hidden-response Hessian minimization law, the same square-root operator core, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family, the same canonical minimal visible-response representative, the same current body-response core, and every currently recorded downstream benchmark-facing consequence follow unchanged. The remaining honest burden therefore moves below the separate centered residual law itself to a deeper derivation or justification of the centered hidden-response residual-source law that forces it.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has already moved one layer below the square-root-operator-core frontier and then one layer below the hidden-response-Hessian frontier itself. The current active-primary theorem shows that the current hidden-response Hessian minimization law is already the canonical least-squares pullback of a smaller centered hidden-response residual law on the fixed displacement fibers.

The next honest question is therefore narrower again. What still more primitive explicit substrate structure forces that centered residual law itself? The present note answers that question in the smallest clean source-law form now available below the current frontier. It does not reinstall the hidden-response Hessian minimization law, the square-root operator core, the normalized-lift package, the selected-lift package, the body-law presentation, the visible-response operator family, the canonical minimal visible-response representative, the current body-response core, or any larger current-image-core or reduced-carrier or projection/body or full substrate-law presentation as primitive data. Instead it keeps only a deeper residual-source carrier, a source-state projection onto the current residual state space, a source displacement readout, and a positive residual-source response operator. The current centered residual law then descends canonically as the quotient minimal-source-energy law of that smaller package.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim a completed first-principles derivation of GR from final substrate microphysics. Its narrower positive role is to show that the current centered hidden-response residual law already follows canonically from a smaller centered hidden-response residual-source law on the fixed displacement fibers.

2. Fixed Inputs from the Current Centered-Residual-Law Frontier

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

   satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

   for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded benchmark base and downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active route already fixes the following data and consequences.*

1. *the current proto-section benchmark base and in particular the recorded proto-section body* $\mathcal{B}_\sigma^{\mathrm{psec}}$ *and shell* $\mathcal{Z}_\sigma^{\mathrm{psec}}$*;*

2. *the fixed section-centered displacement body*

$$\mathcal{D}_\sigma^{\mathrm{ppprese c}} \subset \mathcal{B}_\sigma^{\mathrm{psec}} \times V_\sigma^{\mathrm{ppprese c}}$$

   whose fiber

$$\mathcal{D}_\sigma^{\mathrm{ppprese c}}(y) \equiv \{v \in V_\sigma^{\mathrm{ppprese c}} : (y, v) \in \mathcal{D}_\sigma^{\mathrm{ppprese c}}\}$$

   is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*;*

3. *the recorded downstream continuation*

$$\mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,bodyresp}}, \quad \mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,bodyop}}, \quad \mathfrak{R}^{\mathrm{sub,proto,prese c,res}}, \quad \mathfrak{R}^{\mathrm{sub,proto,prese c}},$$
$$\mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb,op}}, \quad \mathfrak{S}^{\mathrm{sub,proto,ppprese c,pb}}, \quad \mathfrak{R}^{\mathrm{sub,proto,prese c,sec}},$$

   together with the current continuity-Hessian compressed core and benchmark one-metric observer package;

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Definition 3 (Benchmark-faithful centered hidden-response residual law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* benchmark-faithful centered hidden-response residual law *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*

2. *A finite-dimensional Euclidean centered residual state space*

$$\mathcal{U}_\sigma^{\mathrm{disp,hid,res}}.$$

3. *A smooth family of surjective linear displacement readouts*

$$\Pi_\sigma^{\mathrm{disp,hid,res}}(y) : \mathcal{U}_\sigma^{\mathrm{disp,hid,res}} \to V_\sigma^{\mathrm{ppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

4. *A finite-dimensional Euclidean residual target*

$$\mathcal{E}_\sigma^{\mathrm{disp,hid,res}}.$$

5. *A smooth family of injective linear centered residual operators*

$$L_\sigma^{\mathrm{disp,hid,res}}(y) : \mathcal{U}_\sigma^{\mathrm{disp,hid,res}} \to \mathcal{E}_\sigma^{\mathrm{disp,hid,res}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

6. *Benchmark faithfulness in the sense of Definition 2(4).*

Proposition 1 (What the current centered-residual frontier already proves)**.** *At current scope, the already-recorded centered-residual frontier theorems prove all of the following.*

1. *Any benchmark-faithful centered hidden-response residual law canonically determines the current hidden-response Hessian minimization law by*

$$\begin{aligned}
\mathcal{H}_\sigma^{\mathrm{disp,hid}} &\equiv \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}, \\
\Pi_\sigma^{\mathrm{disp,hid}}(y) &\equiv \Pi_\sigma^{\mathrm{disp,hid,res}}(y), \\
A_\sigma^{\mathrm{disp,hid}}(y) &\equiv L_\sigma^{\mathrm{disp,hid,res}}(y)^* L_\sigma^{\mathrm{disp,hid,res}}(y).
\end{aligned}$$

2. *That hidden-response Hessian minimization law canonically determines the current square-root operator core on the fixed displacement fibers.*

3. *That square-root operator core canonically determines the current section-centered displacement-body quadratic forcing law, the visible-response operator family $S_\sigma^{\mathrm{disp,resp,vis}}$, and the canonical minimal visible-response representative $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$.*

4. *The current body-response core and every currently recorded downstream benchmark-facing consequence then follow unchanged.*

Proof. Item (1) is exactly the content of the current theorem proving that the hidden-response Hessian minimization law is already the canonical least-squares pullback of a centered hidden-response residual law. Items (2)–(4) are exactly the already-recorded downstream continuation from that hidden-response-Hessian frontier. □

3. Centered Hidden-Response Residual-Source Law

Definition 4 (Benchmark-faithful centered hidden-response residual-source law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* benchmark-faithful centered hidden-response residual-source law *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*
2. *A finite-dimensional Euclidean residual-source carrier*
$$\mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}.$$
3. *A smooth family of surjective linear source-state projections*
$$Q_\sigma^{\mathrm{disp,hid,res,src}}(y) : \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}} \to \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$
4. *A smooth family of surjective linear source displacement readouts*
$$\Theta_\sigma^{\mathrm{disp,hid,res,src}}(y) : \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}} \to V_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$
5. *Exactness of the displacement readout on source-state fibers:*
$$\ker Q_\sigma^{\mathrm{disp,hid,res,src}}(y) \subseteq \ker \Theta_\sigma^{\mathrm{disp,hid,res,src}}(y) \qquad \textit{for every } y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$
6. *A smooth family of positive self-adjoint residual-source response operators*
$$K_\sigma^{\mathrm{disp,hid,res,src}}(y) : \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}} \to \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$
each of which is positive definite on $\mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}$.
7. *Benchmark faithfulness in the sense of Definition 2(4).*

Remark 1. *Definition 4 is intentionally smaller at benchmark scope than the current centered residual law. It does not install a separate residual target, a separate residual operator, or even a separately primitive residual displacement readout. Instead the residual state space appears as the quotient output of a deeper source carrier through* $Q_\sigma^{\mathrm{disp,hid,res,src}}(y)$*, the residual readout descends from* $\Theta_\sigma^{\mathrm{disp,hid,res,src}}(y)$*, and the residual operator will be induced by minimal source energy. It also does not reinstall the hidden-response Hessian minimization law, the square-root operator core, the normalized-lift package, the body-level forcing law, the visible-response operator family, the canonical minimal visible-response representative, the current body-response core, or any larger same-output presentation as primitive data.*

4. Induced Centered Hidden-Response Residual Law

Definition 5 (Minimal residual-source lift)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *and a benchmark-faithful centered hidden-response residual-source law. For* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*, define the residual-source quotient metric*
$$M_\sigma^{\mathrm{disp,hid,res,src}}(y) \equiv Q_\sigma^{\mathrm{disp,hid,res,src}}(y) K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1} Q_\sigma^{\mathrm{disp,hid,res,src}}(y)^*. \tag{2}$$

Also define the minimal residual-source lift by

$$J_\sigma^{\mathrm{disp,hid,res,src}}(y) \equiv K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1} Q_\sigma^{\mathrm{disp,hid,res,src}}(y)^* M_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1} : \mathcal{U}_\sigma^{\mathrm{disp,hid,res}} \to \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}. \tag{3}$$

Finally define the residual-source energy

$$\mathcal{E}_\sigma^{\mathrm{disp,hid,res,src}}(y, w) \equiv \frac{1}{2} \left\langle w, K_\sigma^{\mathrm{disp,hid,res,src}}(y) w \right\rangle, \qquad w \in \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}.$$

Proposition 2 (The residual-source quotient metric is positive definite)**.** *For every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$, the operator $M_\sigma^{\mathrm{disp,hid,res,src}}(y)$ of (2) is positive self-adjoint and positive definite on $\mathcal{U}_\sigma^{\mathrm{disp,hid,res}}$.*

Proof. Self-adjointness is immediate from the definition. For any $u \in \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}$,

$$\left\langle u, M_\sigma^{\mathrm{disp,hid,res,src}}(y) u \right\rangle = \left\langle K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1/2} Q_\sigma^{\mathrm{disp,hid,res,src}}(y)^* u, K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1/2} Q_\sigma^{\mathrm{disp,hid,res,src}}(y)^* u \right\rangle.$$

The right-hand side is nonnegative and vanishes only if

$$Q_\sigma^{\mathrm{disp,hid,res,src}}(y)^* u = 0.$$

Because $Q_\sigma^{\mathrm{disp,hid,res,src}}(y)$ is surjective, its adjoint is injective, so $Q_\sigma^{\mathrm{disp,hid,res,src}}(y)^* u = 0$ implies $u = 0$. Hence $M_\sigma^{\mathrm{disp,hid,res,src}}(y)$ is positive definite. □

Proposition 3 (Minimal-source-energy characterization)**.** *For every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$, the map $J_\sigma^{\mathrm{disp,hid,res,src}}(y)$ of (3) is the unique linear right inverse of $Q_\sigma^{\mathrm{disp,hid,res,src}}(y)$ whose value at $u \in \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}$ minimizes the residual-source energy on the affine fiber*

$$\left\{ w \in \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}} : Q_\sigma^{\mathrm{disp,hid,res,src}}(y) w = u \right\}.$$

More precisely,

$$Q_\sigma^{\mathrm{disp,hid,res,src}}(y) J_\sigma^{\mathrm{disp,hid,res,src}}(y) = \mathrm{id}_{\mathcal{U}_\sigma^{\mathrm{disp,hid,res}}}, \tag{4}$$

and for every $u \in \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}$,

$$\mathcal{E}_\sigma^{\mathrm{disp,hid,res,src}}(y, J_\sigma^{\mathrm{disp,hid,res,src}}(y) u) = \min_{Q_\sigma^{\mathrm{disp,hid,res,src}}(y) w = u} \mathcal{E}_\sigma^{\mathrm{disp,hid,res,src}}(y, w). \tag{5}$$

Proof. Equation (4) follows from Definitions 5 and Proposition 2:

$$\begin{aligned} Q_\sigma^{\mathrm{disp,hid,res,src}}(y) J_\sigma^{\mathrm{disp,hid,res,src}}(y) &= Q_\sigma^{\mathrm{disp,hid,res,src}}(y) K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1} Q_\sigma^{\mathrm{disp,hid,res,src}}(y)^* M_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1} \\ &= M_\sigma^{\mathrm{disp,hid,res,src}}(y) M_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1} = \mathrm{id}. \end{aligned}$$

Fix $u \in \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}$ and write

$$w = J_\sigma^{\mathrm{disp,hid,res,src}}(y) u + \eta, \qquad \eta \in \ker Q_\sigma^{\mathrm{disp,hid,res,src}}(y).$$

Because $J_\sigma^{\mathrm{disp,hid,res,src}}(y)$ is built from the Euler-Lagrange formula for the constrained quadratic

problem, we have

$$\begin{aligned}\left\langle K_\sigma^{\mathrm{disp,hid,res,src}}(y) J_\sigma^{\mathrm{disp,hid,res,src}}(y) u, \eta \right\rangle &= \left\langle Q_\sigma^{\mathrm{disp,hid,res,src}}(y)^* M_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1} u, \eta \right\rangle \\ &= \left\langle M_\sigma^{\mathrm{disp,hid,res,src}}(y)^{-1} u, Q_\sigma^{\mathrm{disp,hid,res,src}}(y) \eta \right\rangle = 0.\end{aligned}$$

Therefore

$$\mathcal{E}_\sigma^{\mathrm{disp,hid,res,src}}(y, w) = \mathcal{E}_\sigma^{\mathrm{disp,hid,res,src}}(y, J_\sigma^{\mathrm{disp,hid,res,src}}(y) u) + \frac{1}{2} \left\langle \eta, K_\sigma^{\mathrm{disp,hid,res,src}}(y) \eta \right\rangle.$$

The second term is nonnegative and vanishes only when $\eta = 0$ because $K_\sigma^{\mathrm{disp,hid,res,src}}(y)$ is positive definite. This proves (5) and uniqueness of the minimizer. □

Proposition 4 (The source displacement readout descends to the residual state space)**.** *Under Definition 4, there is a unique smooth family of surjective linear maps*

$$\Pi_\sigma^{\mathrm{disp,hid,res}}(y) : \mathcal{U}_\sigma^{\mathrm{disp,hid,res}} \to V_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

such that

$$\Pi_\sigma^{\mathrm{disp,hid,res}}(y) Q_\sigma^{\mathrm{disp,hid,res,src}}(y) = \Theta_\sigma^{\mathrm{disp,hid,res,src}}(y). \tag{6}$$

Equivalently,

$$\Pi_\sigma^{\mathrm{disp,hid,res}}(y) = \Theta_\sigma^{\mathrm{disp,hid,res,src}}(y) J_\sigma^{\mathrm{disp,hid,res,src}}(y). \tag{7}$$

Proof. The inclusion

$$\ker Q_\sigma^{\mathrm{disp,hid,res,src}}(y) \subseteq \ker \Theta_\sigma^{\mathrm{disp,hid,res,src}}(y)$$

means that $\Theta_\sigma^{\mathrm{disp,hid,res,src}}(y)$ is constant on the fibers of $Q_\sigma^{\mathrm{disp,hid,res,src}}(y)$. Hence a unique descended map $\Pi_\sigma^{\mathrm{disp,hid,res}}(y)$ exists satisfying (6). Surjectivity of $\Pi_\sigma^{\mathrm{disp,hid,res}}(y)$ follows from surjectivity of $\Theta_\sigma^{\mathrm{disp,hid,res,src}}(y)$ and $Q_\sigma^{\mathrm{disp,hid,res,src}}(y)$. Equation (7) is immediate from (6) and (4). Smoothness follows because both source maps and $J_\sigma^{\mathrm{disp,hid,res,src}}(y)$ vary smoothly in y. □

Definition 6 (Residual-source-induced centered residual operator)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ and a benchmark-faithful centered hidden-response residual-source law. Define the induced residual target and residual operator by*

$$\mathcal{E}_\sigma^{\mathrm{disp,hid,res}} \equiv \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}, \qquad L_\sigma^{\mathrm{disp,hid,res}}(y) \equiv K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{1/2} J_\sigma^{\mathrm{disp,hid,res,src}}(y).$$

Proposition 5 (The induced centered residual operator is injective)**.** *Under Definition 6, each $L_\sigma^{\mathrm{disp,hid,res}}(y)$ is injective.*

Proof. If $L_\sigma^{\mathrm{disp,hid,res}}(y) u = 0$, then

$$K_\sigma^{\mathrm{disp,hid,res,src}}(y)^{1/2} J_\sigma^{\mathrm{disp,hid,res,src}}(y) u = 0.$$

Because $K_\sigma^{\mathrm{disp,hid,res,src}}(y)$ is positive definite, its square root is invertible, so $J_\sigma^{\mathrm{disp,hid,res,src}}(y) u = 0$. Applying (4) yields

$$u = Q_\sigma^{\mathrm{disp,hid,res,src}}(y) J_\sigma^{\mathrm{disp,hid,res,src}}(y) u = 0.$$

Hence $L_\sigma^{\mathrm{disp,hid,res}}(y)$ is injective. □

Proposition 6 (The induced residual operator is exact minimal-source energy)**.** *Under Definition 6, for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ and every $u \in \mathcal{U}_\sigma^{\mathrm{disp,hid,res}}$,*

$$\frac{1}{2}\left\|L_\sigma^{\mathrm{disp,hid,res}}(y)u\right\|^2 = \mathcal{E}_\sigma^{\mathrm{disp,hid,res,src}}\big(y, J_\sigma^{\mathrm{disp,hid,res,src}}(y)u\big) = \min_{Q_\sigma^{\mathrm{disp,hid,res,src}}(y)w=u} \mathcal{E}_\sigma^{\mathrm{disp,hid,res,src}}(y,w). \tag{8}$$

Proof. By Definition 6,

$$\begin{aligned}\frac{1}{2}\left\|L_\sigma^{\mathrm{disp,hid,res}}(y)u\right\|^2 &= \frac{1}{2}\left\langle J_\sigma^{\mathrm{disp,hid,res,src}}(y)u, K_\sigma^{\mathrm{disp,hid,res,src}}(y)J_\sigma^{\mathrm{disp,hid,res,src}}(y)u\right\rangle \\ &= \mathcal{E}_\sigma^{\mathrm{disp,hid,res,src}}\big(y, J_\sigma^{\mathrm{disp,hid,res,src}}(y)u\big).\end{aligned}$$

The second equality in (8) is exactly Proposition 3. □

Proposition 7 (Residual-source-induced centered hidden-response residual law)**.** *Under Definitions 4 and 6, the data*

$$\left(\mathcal{U}_\sigma^{\mathrm{disp,hid,res}}, \Pi_\sigma^{\mathrm{disp,hid,res}}, \mathcal{E}_\sigma^{\mathrm{disp,hid,res}}, L_\sigma^{\mathrm{disp,hid,res}}\right)$$

form a benchmark-faithful centered hidden-response residual law in the sense of Definition 3.

Proof. The state space $\mathcal{U}_\sigma^{\mathrm{disp,hid,res}}$ is finite-dimensional and Euclidean by Definition 3. Proposition 4 gives a smooth family of surjective residual readouts. The target $\mathcal{E}_\sigma^{\mathrm{disp,hid,res}} = \mathcal{W}_\sigma^{\mathrm{disp,hid,res,src}}$ is finite-dimensional and Euclidean. Proposition 5 shows that the induced residual operator is injective. Benchmark faithfulness is inherited directly from Definition 4(7). Therefore all items of Definition 3 hold. □

5. Deriving the Current Centered-Residual Frontier

Theorem 1 (Current centered residual law from a centered hidden-response residual-source law)**.** *Fix the primitive continuation data of Definition 1, the recorded benchmark base and downstream chain of Definition 2, and for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ a benchmark-faithful centered hidden-response residual-source law in the sense of Definition 4. Then the current centered hidden-response residual law is canonically forced by that smaller explicit substrate package. More precisely:*

1. *Propositions 4 and 7 canonically associate to the residual-source law a benchmark-faithful centered hidden-response residual law $(\mathcal{U}_\sigma^{\mathrm{disp,hid,res}}, \Pi_\sigma^{\mathrm{disp,hid,res}}, \mathcal{E}_\sigma^{\mathrm{disp,hid,res}}, L_\sigma^{\mathrm{disp,hid,res}})$.*

2. *Proposition 6 identifies the current residual operator exactly with the quotient minimal-source-energy law*

$$\frac{1}{2}\left\|L_\sigma^{\mathrm{disp,hid,res}}(y)u\right\|^2 = \min_{Q_\sigma^{\mathrm{disp,hid,res,src}}(y)w=u} \frac{1}{2}\left\langle w, K_\sigma^{\mathrm{disp,hid,res,src}}(y)w\right\rangle.$$

3. *By Proposition 1, the current hidden-response Hessian minimization law, the same square-root operator core, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family $S_\sigma^{\mathrm{disp,resp,vis}}$, the same canonical minimal visible-response representative $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$, the same current body-response core, and every currently recorded downstream benchmark-facing consequence then follow unchanged from that same centered residual-source law through the induced centered residual law.*

4. *The benchmark one-metric package* (1) *is preserved, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

5. *Consequently the remaining honest burden is no longer to restate the current centered hidden-response residual law itself, but to derive or justify the centered hidden-response residual-source law that forces it from still more primitive explicit substrate structure.*

Proof. Item (1) is Propositions 4 and 7. Item (2) is Proposition 6. Item (3) is the already-recorded downstream continuation of the current centered-residual frontier applied to the law furnished by Items (1)–(2). Item (4) is the benchmark-faithfulness condition built into Definitions 2 and 4. Item (5) is the routing consequence of Items (1)–(4): once the current centered residual law is shown to be the canonical quotient minimal-source-energy law of the smaller centered residual-source package, another manuscript that merely restates that same residual law or merely replays the same hidden-response-Hessian or square-root-core or normalized-lift or selected-lift or body-law or visible-response consequences does not remove any remaining benchmark-facing freedom. □

Corollary 1 (The centered hidden-response residual law is no longer primitive)**.** *Under the hypotheses of Theorem 1, the current centered hidden-response residual law is no longer a primitive stopping point on the recorded section-centered hidden-response route. Its benchmark-relevant content is already forced by the smaller centered hidden-response residual-source law* $\mathfrak{P}^{\mathrm{disp,hid,res,src}}$.

Proof. Theorem 1(1)–(3) identifies the current centered residual law and its same hidden-response-Hessian and downstream benchmark-facing consequences as outputs of the smaller centered residual-source law. Therefore the current centered-residual package is no longer primitive at benchmark scope once that source law is fixed. □

6. Conclusion

The positive result of this note is a genuine derivational gain below the current centered-residual frontier. The current centered hidden-response residual law no longer needs to be treated as primitive. It is already the canonical quotient minimal-source-energy law of a smaller centered hidden-response residual-source law on the fixed displacement fibers.

The scope remains narrow and honest. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and nothing here upgrades adoption to a completed first-principles derivation of GR from final substrate microphysics.

The next open burden is therefore sharper again. The next benchmark-facing manuscript should derive or justify the centered hidden-response residual-source law itself from still more primitive explicit substrate structure. Another manuscript that merely restates the same centered residual-source law, the same centered residual law, the same hidden-response Hessian minimization law, the same square-root operator core, the same carrier-bearing normalized selected-response lift, the same operator-bearing selected-lift package, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family, the same canonical minimal visible-response representative, the same current body-response core, or a larger same-output current-image-core or reduced-carrier or projection/body or full substrate-law presentation is no longer the clean next target at current scope.